

Pass-Through

C.Conlon

Fall 2026

Grad IO

What is pass-through and why do we care?

Pass-Through: What is it?

Lots of cases in economics where we want to know how prices respond to changes in costs:

- ▶ Response to cost shocks (oil shock, commodity prices)
- ▶ Response to tax changes (incidence/efficiency of taxes)
- ▶ Transmission of exchange rate shocks
- ▶ Transmission of monetary policy
- ▶ Price effects of mergers: converting **UPP** (a change in opportunity cost) into a change in price
- ▶ Double marginalization: how much of a wholesale price change reaches the shelf

Today: what determines pass-through in theory, what the demand models of this course imply for it, and why it is so hard to estimate.

Pass-through is the simplest thing in economics that nobody understands

► Often we are talking about different things

- IO: mostly $\rho = \frac{\partial p_j}{\partial mc_j}$ or the matrix $\frac{\partial \mathbf{p}}{\partial \mathbf{mc}}$
- Macro/Trade: mostly $\frac{\partial \log p_j}{\partial \log mc_j}$
- (Throughout this lecture ρ is the **pass-through rate** – not the nesting parameter from the demand lectures.)

► One thing we know for sure: constant marginal costs, perfect competition:

$$p = mc \rightarrow \frac{\partial p}{\partial mc} = 1$$

► Everything else depends on two things:

- **Curvature** of demand (second derivatives)
- Nature of **competition** (conduct)

► Neither is something an own-price elasticity pins down. That is the whole lecture.

The Empirical Puzzle: Estimates Are All Over the Place

Shock → price	$\hat{\rho}$	Source
Exchange rate → US import prices, dollar-priced	0.25	Gopinath Itskhoki Rigobon 2010
same, priced in foreign currency	0.95	
Exchange rate → retail beer	≈ 0.25	Goldberg Hellerstein 2013
Commodity (coffee) → retail	≈ 0.30	Nakamura Zerom 2010
Cigarette taxes	< 1 / ≈ 1	Harding et al. 2012 / DeCicca et al. 2013
Gasoline taxes	≈ 1 ; holidays $\ll 1$	Marion Muehlegger 2011; Doyle Samphantharak 2008
Sales taxes	≈ 1 , half > 1	Poterba 1996; Besley Rosen 1999
Alcohol taxes	1.5 – 2.1	Young Bielinska-Kwapisz 2002; Kenkel 2005

- ▶ Trade theory (CES + monopolistic competition) predicts 0 or 1 depending on invoicing currency. The data say 0.25 and 0.95.
- ▶ Anything from 0.25 to 2 shows up. Theory says that is **exactly what to expect**: pass-through is about curvature and conduct, and both vary across markets.
- ▶ Many “price elasticity” studies (gasoline, cigarettes, alcohol) assume $\rho = 1$ and regress quantities on tax changes: $\frac{\partial \log q}{\partial \log t} = \frac{\partial \log q}{\partial \log p}$ (!)

One product, one price

Perfect Competition: Jenkin (1872), Alfred Marshall (1890)

$$D(p) = S(p - t)$$
$$\rho = \frac{dp}{dt} = \frac{1}{1 + \frac{\varepsilon_d}{\varepsilon_s}}$$

- ▶ Where ε_d is demand elasticity and ε_s is supply elasticity respectively.
- ▶ $\rho \in [0, 1]$ as long demand slopes down and supply slopes up!
- ▶ Constant MC implies that $\varepsilon_s \rightarrow \infty$ and $\rho \rightarrow 1$
- ▶ Incidence $I = \frac{\frac{dCS}{dt}}{\frac{dPS}{dt}} = \frac{\rho}{1-\rho}$ (higher pass-through, more borne by consumers)

So far so good. Nothing about curvature yet: with perfect competition only slopes matter.

Monopoly (Weyl and Fabinger, JPE 2013)

Start with $MR = MC$ and differentiate with respect to the tax t :

$$p(q) + p'(q)q = mc(q) + t \Rightarrow \frac{dq}{dt} = \frac{1}{mr' - mc'} \Rightarrow \rho = \frac{dp}{dt} = p' \frac{dq}{dt} = \frac{p'}{mr' - mc'}$$

Define the markup term $ms(q) \equiv -p'(q)q$ and the elasticities $\epsilon_D = -\frac{p}{p'q}$, $\epsilon_S = \frac{mc}{mc'q}$, $\epsilon_{ms} = \frac{ms}{ms'q}$. Using the Lerner index $\frac{mc}{p} = \frac{\epsilon_D - 1}{\epsilon_D}$:

$$\boxed{\rho = \frac{1}{1 + \frac{\epsilon_D - 1}{\epsilon_S} + \frac{1}{\epsilon_{ms}}}} \quad \text{constant MC: } \rho = \frac{1}{1 + 1/\epsilon_{ms}}, \quad \frac{1}{\epsilon_{ms}} = 1 + \frac{p''q}{p'}$$

- ▶ $\epsilon_D > 1$ for a monopolist (except for my MBA students).
- ▶ $1/\epsilon_{ms}$ is the **curvature** of (inverse) demand, and $(\log D)'' = -\frac{1}{\epsilon_{ms}} \frac{1}{ms^2}$, so its sign is the sign of the curvature of **log demand**:
 - log-concave demand $\Rightarrow 1/\epsilon_{ms} > 0 \Rightarrow \rho < 1$ (linear demand: $\rho = 1/2$)
 - log-linear (exponential) demand $\Rightarrow \rho = 1$
 - log-convex demand $\Rightarrow \rho > 1$ (constant elasticity)
- ▶ Bulow and Pfleiderer (1983): with constant MC a monopolist's pass-through is **entirely** about curvature.

Canonical Demand Curves (single product, constant MC)

Demand	$\rho = dp/dc$	$\log D$	Markup
Perfect competition	1	–	0
Linear $q = a - bp$	1/2	concave	$(p - c) = \frac{a-bc}{2b}$
Exponential $q = Ae^{-\lambda p}$	1	linear	$p - c = 1/\lambda$ (constant)
Constant elasticity $q = Ap^{-\epsilon}$	$\frac{\epsilon}{\epsilon-1} > 1$	convex	$p = \frac{\epsilon}{\epsilon-1} c$ (constant Lerner)
Logit, single-product firm, rivals fixed	$1 - \sigma_j$	concave	$p_j - c_j = \frac{1}{\alpha(1-\sigma_j)}$

- ▶ Linear, exponential and constant-elasticity are the $\rho = \frac{1}{2}, 1, > 1$ members of the Bulow–Pfleiderer (1983) constant pass-through family; the puzzle slide lives between the linear and CES rows.
- ▶ A **constant absolute markup** (exponential) gives $\rho = 1$; a **constant percentage markup** (CES) gives $\rho > 1$ because the markup scales with cost. Hence CES trade models over-shift unless pricing-to-market or invoicing is added.
- ▶ Logit sits below one and gets **closer** to one as the product gets smaller. Derivation in two slides.

Conduct and Incidence (Weyl and Fabinger 2013)

Generalize to symmetric imperfect competition with a conduct parameter θ : $\theta = 0$ (PC), $\theta = 1$ (monopoly), $\theta = 1/N$ (symmetric Cournot):

$$\rho = \frac{1}{1 + \frac{\theta}{\epsilon_{\theta}} + \frac{\epsilon_D - \theta}{\epsilon_S} + \frac{\theta}{\epsilon_{ms}}} \quad \text{with } \epsilon_{\theta} \equiv \frac{\theta}{\theta' q} \text{ (how conduct itself changes with quantity).}$$

Incidence follows from pass-through and conduct alone (their “Principle of Incidence”):

$$\frac{1}{I} = \frac{1}{\rho} - (1 - \theta) \iff I = \frac{\rho}{1 - (1 - \theta)\rho} : \quad I_{PC} = \frac{\rho}{1 - \rho}, \quad I_{\text{monopoly}} = \rho.$$

- Monopoly: producer surplus falls by exactly q per dollar of tax (envelope theorem), consumer surplus by ρq . So $\rho > 1$ means consumers bear **more than the whole tax**.
- Two sufficient statistics for incidence, (ρ, θ) , neither of which is an elasticity.
- Not sure how I feel about the revival of the conduct parameter (more in the conduct lecture).

Elasticity vs. Curvature: Mrázová and Neary (AER 2017)

Write everything in terms of inverse demand $p(x)$. Two local statistics at any point:

$$\varepsilon(x) \equiv -\frac{p(x)}{x p'(x)} > 0 \quad (\text{elasticity}) \qquad \kappa(x) \equiv -\frac{x p''(x)}{p'(x)} \quad (\text{convexity; MN call it } \rho)$$

$$\text{Monopoly, constant MC:} \quad \frac{dp}{dc} = \frac{1}{2 - \kappa} \quad \Rightarrow \quad \frac{dp}{dc} - 1 = \frac{\kappa - 1}{2 - \kappa}$$

- ▶ Second-order condition: $\kappa < 2$. Over-shifting ($\rho > 1$) iff $\kappa > 1$: demand more convex than exponential.
- ▶ **Superconvex** demand ($\kappa > 1 + 1/\varepsilon$, more convex than CES at the same elasticity): the proportional markup p/c rises with cost. CES is the boundary.
- ▶ The **demand manifold**: any demand function traces a curve in (κ, ε) space. The elasticity sets the markup, the convexity sets the pass-through, and **knowing one tells you nothing about the other**.
- ▶ Our problem in one line: demand estimation targets ε ; cost-change counterfactuals need κ , which the functional form supplies for free.

Mrázová and Neary (2017): The Space of Elasticity and Convexity

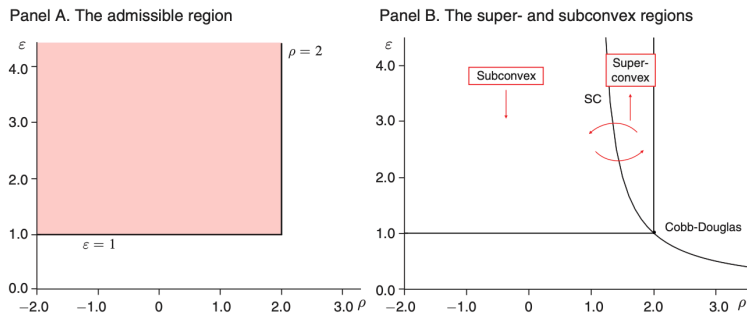


FIGURE 1. THE SPACE OF ELASTICITY AND CONVEXITY

Panel A: the admissible region ($\varepsilon > 1, \kappa < 2$). Panel B: the SC curve is the CES family (ε constant); below it demand is subconvex (proportional markups fall with cost), above it superconvex. Cobb–Douglas is the corner.

Discrete choice

Logit Pass-Through in Two Lines

Single-product firm, plain logit, rivals' prices held fixed. The FOC is $p_j = c_j + \eta_j(p_j)$ with

$$\eta_j = \frac{1}{\alpha(1 - \sigma_j)}, \quad \frac{\partial \eta_j}{\partial p_j} = \frac{1}{\alpha} \frac{1}{(1 - \sigma_j)^2} \frac{\partial \sigma_j}{\partial p_j} = -\frac{\sigma_j}{1 - \sigma_j} \quad \Rightarrow \quad \rho_j = \frac{1}{1 - \partial \eta_j / \partial p_j} = \boxed{1 - \sigma_j}$$

- ▶ Logit demand is **log-concave** in own price: $(\log \sigma_j)'' = -\alpha^2 \sigma_j (1 - \sigma_j) < 0$, so $\rho_j < 1$ always.
- ▶ The markup $\frac{1}{\alpha(1 - \sigma_j)}$ *falls* as the price rises (share shrinks), which is what absorbs part of the cost increase. A niche product passes through almost everything; a dominant product very little.
- ▶ Note what is **not** in the formula: α . The price coefficient sets the level of markups and elasticities; the curvature, and hence pass-through, is pinned down by the **share alone**. This is the second-derivative analogue of IIA.
- ▶ With rivals responding (strategic complements) own pass-through is higher and cross pass-through is positive: that needs the matrix (next section). The $1 - \sigma_j$ is its “rivals fixed” diagonal.

Mixed Logit: Curvature Comes From the Mixing Distribution

Aggregate demand is a mixture, $\sigma_j(p) = \sum_i w_i \sigma_{ij}(p)$, and mixtures of log-concave functions need not be log-concave.

- ▶ Raise p_j : the price-sensitive types (high α_i) leave first, so the **residual** demand is less elastic than the average. That selection makes $\log \sigma_j$ more convex than any single type's, and $\rho_j > 1$ becomes possible.
- ▶ So in a BLP model pass-through is a byproduct of the **distribution you assumed for α_i** (and the characteristics with random coefficients), which estimation on prices and shares barely disciplines. Normal, lognormal, or a two-point α_i can all fit the same elasticities and imply different pass-through.
- ▶ Miravete, Seim, Thurk (2023, "Elasticity and Curvature of Discrete Choice Demand Models"): map the (ε, κ) pairs each specification can reach in equilibrium. Plain logit and nested logit are a curve; mixed logit is a region whose shape is set by the mixing family, not the data.
- ▶ Practical advice: (a) report the implied pass-through next to the elasticities, it is a cheap diagnostic; (b) if you have cost or tax variation, **use it as a supply moment** (Conlon Rao in the BLP lecture matched observed markups and a tax change for exactly this reason); (c) if your counterfactual is a cost change, the mixing distribution is doing the work, so vary it.

Many products: the pass-through matrix

Multiproduct Bertrand: Pass-Through Is a Matrix

Recall the FOC in markup form from the merger simulation lecture, $\Omega(\mathbf{p}) \equiv \mathcal{H} \odot \Delta(\mathbf{p})$:

$$f(\mathbf{p}, \mathbf{mc}) \equiv \mathbf{p} - \mathbf{mc} - \underbrace{\Omega(\mathbf{p})^{-1} \boldsymbol{\sigma}(\mathbf{p})}_{\boldsymbol{\eta}(\mathbf{p})} = 0.$$

The implicit function theorem, with $\partial f / \partial \mathbf{mc} = -\mathbb{I}_J$ because costs enter additively:

$$\mathcal{P} \equiv \frac{\partial \mathbf{p}}{\partial \mathbf{mc}} = - \left(\frac{\partial f}{\partial \mathbf{p}} \right)^{-1} \frac{\partial f}{\partial \mathbf{mc}} = \left(\mathbb{I}_J - \frac{\partial \boldsymbol{\eta}}{\partial \mathbf{p}} \right)^{-1}, \quad \mathcal{P}_{(j,k)} = \frac{\partial p_j}{\partial mc_k}.$$

- ▶ Diagonal: own-cost pass-through, now including rivals' equilibrium responses. Off-diagonal: a rival's cost shock moves my price (positive under strategic complements).
- ▶ Single-product logit: $\partial \eta_j / \partial p_j = -\sigma_j / (1 - \sigma_j)$ on the diagonal; ignoring the off-diagonal terms gives back $1 - \sigma_j$.
- ▶ Multiproduct firms: η_j depends on the margins of all sister products, so a cost shock to k moves p_j with no rival response at all.
- ▶ Everything hard is in $\partial \boldsymbol{\eta} / \partial \mathbf{p}$: the derivative of a markup is a **second derivative of demand**.

How do we get the pass-through matrix?

Two ways:

1. Numerically: perturb the marginal cost of each product k in turn and re-solve for equilibrium prices

$$\mathbf{p} = \mathbf{mc} + e_k + \Omega(\mathbf{p})^{-1} \boldsymbol{\sigma}(\mathbf{p})$$

(Morrow Skerlos (2011) fixed point, J solves, finite differences; fine for small J .)

2. Analytic: differentiate the FOC and apply the implicit function theorem (next slide).

See Jaffe Weyl (AEJM 2013), Miller Weinberg (2017, Appendix E), Conlon Rao (2025).

This is what PyBLP does with `results.compute_passthrough`, building the full $J \times J \times J$ demand Hessian (slow, and J^3 memory). JaxBLP contracts that tensor with the markups analytically first and never forms it ($J^2 I$ flops; 0.4s vs. 3.7s and 31GB on a 636-product market).

Multivariate IFT: The Hard Part

Differentiate the FOC with respect to p_l to get column l of $\partial f / \partial \mathbf{p}$:

$$\frac{\partial f(\mathbf{p}, \mathbf{mc})}{\partial p_l} \equiv e_l + \Omega^{-1}(\mathbf{p}) \left[\mathcal{H} \odot \frac{\partial \Delta(\mathbf{p})}{\partial p_l} \right] \Omega^{-1}(\mathbf{p}) \boldsymbol{\sigma}(\mathbf{p}) - \Omega^{-1}(\mathbf{p}) \frac{\partial \boldsymbol{\sigma}(\mathbf{p})}{\partial p_l}. \quad (1)$$

The complicated piece is the demand Hessian: a $J \times J \times J$ tensor with elements (j, k, l) , $\frac{\partial^2 \sigma_j}{\partial p_k \partial p_l}$, since $\frac{\partial \Delta(\mathbf{p})}{\partial p_l} = -\frac{\partial^2 \boldsymbol{\sigma}}{\partial \mathbf{p} \partial p_l}$.

- Compactly, with markups $\boldsymbol{\eta} = \Omega^{-1} \boldsymbol{\sigma}$: $\frac{\partial f}{\partial \mathbf{p}} = \mathbb{I}_J + \Omega^{-1} \left[T - \frac{\partial \boldsymbol{\sigma}}{\partial \mathbf{p}} \right]$ with $T_{(j,l)} = \sum_m \frac{\partial \Omega_{(j,m)}}{\partial p_l} \eta_m$. The tensor only ever appears **contracted with the markups**; computing T directly is what makes the closed form cheap.
- Orientation (as in the pricing lecture): with product-specific α_{ik} , read $\Omega = \mathcal{H} \odot \Delta'$.
- For (mixed) logit the Hessian is a sum of products of σ_{ij} 's and α_i 's: the “curvature from mixing” slide, in matrix form.

Application: From UPP to Price Effects (Jaffe Weyl 2013)

Recall UPP from the antitrust deck: $UPP_j = \Delta c_j + \sum_{k \in \mathcal{J}_g} (p_k - c_k) \cdot D_{jk}(\mathbf{p})$ is a change in **opportunity cost**. But how much do we expect prices to rise?

- ▶ Treat the merger as a cost shock and pass it through: $\Delta \mathbf{p} \approx \mathcal{P} \cdot \mathbf{UPP}$. The first-order approach to merger analysis.
- ▶ That needs the whole $J \times J$ matrix $\mathcal{P}$, which is J^2 numbers (again). So people restrict it: diagonal, or $\mathcal{P} = 0.8 \cdot \mathbb{I}_J$, or just the merging firms' own-cost pass-through.
- ▶ Miller, Remer, Ryan, Sheu (2016) simulate mergers under several demand systems and ask how good the approximation is with the full matrix, the diagonal, or only the merging firms' rows (next slide).
- ▶ Which approximations work depends on demand curvature, and curvature is exactly what the approximation is trying to avoid estimating.

Miller, Remer, Ryan, Sheu (2016): How Good Is the First-Order Approximation?

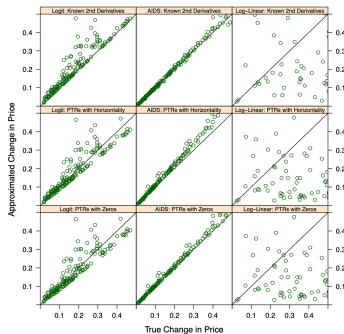


Figure 2: Prediction Error with Complete Information.

Notes: The figure provides scatter-plots of approximation against the true price effect for logit demand, the AIDS and log-linear demand. The case of linear demand is omitted because approximation is exact in that setting. Approximations are calculated alternately based on full knowledge of the second-order properties of demand in neighborhood of pre-merger equilibrium ("Known 2nd Derivatives"); based on full knowledge of pre-merger cost pass-through and the horizontality assumption ("PTRs with Horizontality"); and based on full knowledge of pre-merger cost pass-through and the assumption that derivatives of the form $\partial^2 Q_i / \partial P_j \partial P_k$ equal zero ("PTRs with Zeros").

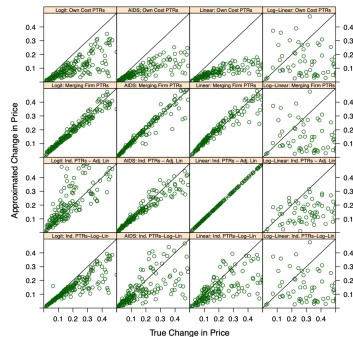


Figure 3: Prediction Error with Incomplete Information

Notes: The figure provides scatter-plots of approximation against the true price effect for logit demand, the AIDS, linear demand and log-linear demand. Four informational scenarios are considered: pre-merger cost pass-through that is available only for own costs ("Own Cost PTRs"); pre-merger cost pass-through that is available only for the merging firms ("Merging Firms PTRs"); industry cost pass-through that is apportioned using the adjusted-linear method ("Ind. PTRs Adj.-Lin."); and industry cost pass-through that is apportioned using the log-linear method ("Ind. PTRs Log-Lin").

Left (complete information): with the full pass-through matrix the approximation is close to exact for AIDS and acceptable for logit; log-linear (CES-like) demand is hopeless because the price effect is far from linear in the cost shock. Right (incomplete information): using only own-cost pass-through understates every price effect; the merging firms' rows do most of the work.

Vertical chains

Double Marginalization: Villas-Boas (ReStud 2007), Miller Weinberg (Ecma 2017)

Most retail prices are set by a retailer who buys from a manufacturer with market power. Retailer and wholesaler FOCs:

$$\mathbf{p}^r = \underbrace{\mathbf{p}^w + \mathbf{c}^r}_{\mathbf{mc}^r} + (\mathcal{H}_r \odot \Delta_r(\mathbf{p}^r))^{-1} \boldsymbol{\sigma}(\mathbf{p}^r)$$
$$\mathbf{p}^w = \mathbf{mc}^w + \left(\mathcal{H}_w \odot \left(\left[\frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w} \right]' \cdot \Delta_r(\mathbf{p}^r) \right) \right)^{-1} \boldsymbol{\sigma}(\mathbf{p}^r)$$

- ▶ Δ_r is the matrix of (retail) demand derivatives with entries $-\frac{\partial \sigma_j}{\partial p_k^r}$ (as in the merger simulation lecture).
- ▶ $\mathcal{H}_r, \mathcal{H}_w$ ownership matrix $(j, k) = 1$ if both products sold by same retailer/wholesaler.
- ▶ $\frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w}$ is the **pass-through matrix** of the retail stage, entry $(m, j) = \partial p_m^r / \partial p_j^w$. Chain rule:
 $-\partial \sigma_k / \partial p_j^w = \sum_m \frac{\partial p_m^r}{\partial p_j^w} \Delta_{r,(m,k)}$, so the wholesaler's matrix is the **transpose** of the pass-through times Δ_r (easy to get wrong in code).

Challenge: We want $\mathbf{p}^r(\mathbf{p}^w)$ and $\mathbf{mc}^w$ but we only have an implicit solution for the retailer FOC. The pass-through matrix is the object that lets the upstream firm's problem be written at all.

Pass-through Counterfactuals? (Conlon Rao 2025)

How do we solve for $\mathbf{p}^w$ under a counterfactual pass-through matrix?

- Idea: pass-through only augments the matrix $\Delta_r(\mathbf{p}^r)$: $\Omega_w \equiv \mathcal{H}_w \odot (\mathcal{P}' \Delta_r)$ with $\mathcal{P} \equiv \frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w}$.
- Example: a constant retail markup or sales tax, $\mathcal{P} = \text{diag}(1 + \tau_r)$, so $\mathcal{P}' = \mathcal{P}$ and $\mathcal{P}' \Delta_r = \mathcal{P} \Lambda - \mathcal{P} \Gamma$ with $\mathcal{P} \Lambda$ still diagonal.

$$\mathbf{p}^w = \mathbf{m} \mathbf{c}^w + \Omega_w^{-1} \boldsymbol{\sigma}(\mathbf{p}^r)$$

Adapt the Morrow Skerlos ζ fixed point (same sign convention as the pricing lecture, $\alpha_i > 0$). For diagonal $\mathcal{P}$, $\mathcal{H}_w \odot (\mathcal{P} \Gamma) = \mathcal{P}(\mathcal{H}_w \odot \Gamma)$, so $\mathcal{P}$ cancels out of the Γ term and survives only on the share term:

$$\mathbf{p}^w \leftrightarrow \mathbf{m} \mathbf{c}^w + \zeta(\mathbf{p}^w) \quad \text{where} \quad \zeta(\mathbf{p}^w) = \Lambda^{-1} [\mathcal{H}_w \odot \Gamma] (\mathbf{p}^w - \mathbf{m} \mathbf{c}^w) + \Lambda^{-1} \mathcal{P}^{-1} \boldsymbol{\sigma}$$

with Λ , Γ , $\boldsymbol{\sigma}$ evaluated at $\mathbf{p}^r(\mathbf{p}^w)$. Higher retail pass-through (bigger $\mathcal{P}$) shrinks the wholesale markup: the wholesaler's own price moves retail prices more than one-for-one.

- General (non-diagonal) $\mathcal{P}$: $\mathcal{P}' \Lambda$ is no longer diagonal, so the clean split fails. Split instead on $D = \text{diag}(\Omega_w)$, iterate $\mathbf{p}^w - \mathbf{m} \mathbf{c}^w \leftarrow D^{-1}(D - \Omega_w)(\mathbf{p}^w - \mathbf{m} \mathbf{c}^w) + D^{-1} \boldsymbol{\sigma}$ with no convergence guarantee, or use Newton with the pass-through Jacobian.

What's the point? Vertical settings

Why do we care about pass-through in vertical settings?

- ▶ The wholesaler's effective demand derivative is $\mathcal{P}' \Delta_r$: with low retail pass-through the manufacturer faces a flatter response and takes a bigger margin; with high pass-through the retailer is doing the marking up.
- ▶ When retail pass-through is high we think that downstream firms have significant market power.
- ▶ This also means the scope for **elimination of double marginalization** is large.
 - Higher pass-through should be associated with more efficient vertical mergers? (Not sure this has been tested)
- ▶ A huge class of models (post-and-hold, minimum markups, RPM, vertical contracts in the second IO course) adds an extra parameter between wholesale and retail prices. All of them are a statement about $\mathcal{P}$.

Taxes and the data

Specific vs. Ad Valorem Taxes: The ζ Map Already Told Us

Producer price $\mathbf{p}$, consumer price $\mathbf{p}^c$. A **specific** tax t adds to cost; an **ad valorem** tax τ scales the consumer price, $\mathbf{p}^c = (1 + \tau)\mathbf{p}$, i.e. $\mathcal{P} = (1 + \tau)\mathbb{I}_J$ on the previous slide:

$$\text{specific: } \mathbf{p} - \mathbf{mc} - t\mathbf{1} = \Lambda^{-1}[\mathcal{H} \odot \Gamma](\mathbf{p} - \mathbf{mc} - t\mathbf{1}) + \Lambda^{-1}\boldsymbol{\sigma}$$

$$\text{ad valorem: } \mathbf{p} - \mathbf{mc} = \Lambda^{-1}[\mathcal{H} \odot \Gamma](\mathbf{p} - \mathbf{mc}) + \frac{1}{1 + \tau}\Lambda^{-1}\boldsymbol{\sigma}$$

- ▶ A specific tax leaves the markup equation alone: the firm re-optimizes against a higher cost and pass-through is ρ from before.
- ▶ An ad valorem tax **taxes the markup**: the share term is scaled by $1/(1 + \tau)$, so the producer gives up markup before any demand response. For the same revenue, ad valorem gives lower consumer prices under market power (Delipalla Keen 1992; Anderson, de Palma, Kreider 2001 for logit oligopoly).
- ▶ Single-product logit, rivals fixed: specific tax $\rho = 1 - \sigma_j$; ad valorem consumer-price pass-through is smaller still.
- ▶ US excise taxes (gasoline, cigarettes, alcohol, the Seattle soda tax) are **specific**; sales taxes are ad valorem. Conlon Rao (2025) compare volume, sales and ethanol taxes on spirits this way.

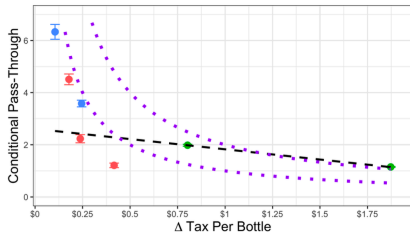
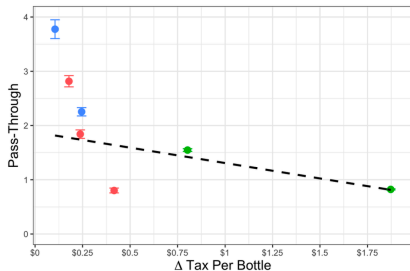
A simple empirical exercise (Conlon Rao, AEJ:EP 2020)

- ▶ US states tax distilled spirits by volume (a specific tax)
- ▶ We see tax changes in three states (Connecticut, Louisiana, Illinois)
- ▶ Three popular bottle sizes (750mL, 1L, 1.75L)
- ▶ Run a regression at a 3 month difference product by product

$$\Delta p_{jst} = \rho_{jst}(\mathbf{X}, \Delta\tau) \cdot \Delta\tau_{jt} + \beta\Delta x_{jst} + \gamma_j + \gamma_t + \epsilon_{jst}$$

- ▶ Only innovation: allow $\rho_{jst}(\mathbf{X}, \Delta\tau)$ to vary by size of tax increase.
- ▶ Prediction from the theory so far: something below one for a logit-like product, around one with constant markups, above one only with very convex demand.

Goes horribly wrong

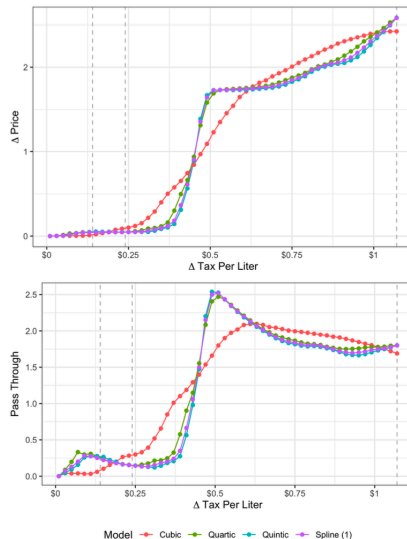


state CT IL LA

- ▶ Regression coefficients are huge > 1 and all over the map
- ▶ Conditional on a price change even larger > 3
- ▶ Seem to coincide with the purple line

Is PT a structural parameter?

- ▶ Price points are a big problem here
- ▶ Estimated an ordered probit with flexible $\beta(\Delta\tau)$
- ▶ Pass-through depends on where you are; not stable at all (!)
- ▶ Maybe structural interpretations of pass-through parameter are a bad idea!
- ▶ The theory is about a smooth FOC. Retail prices move in discrete steps (\$X.99), so a 30 cent tax either does nothing or moves the price a dollar. Averaged over products that can look like anything.



Is It Chain Pricing? DellaVigna Gentzkow (QJE 2019), Butters Sacks Seo (AER 2022)

- ▶ DellaVigna and Gentzkow (2019): US retail chains set nearly **uniform prices** across their stores despite large differences in local demand and income. So a natural excuse for strange local tax pass-through: the chain prices nationally, the local tax is absorbed or spread across all stores.
- ▶ Butters, Sacks, Seo (2022) test it directly: 68 local excise tax changes (beer, liquor, cigarettes, soda) and 76 sales tax changes, 35,151 stores in 96 multi-state chains.
 - Local cost shocks are passed through to **local** prices, at similar rates for national and local chains.
 - **No spillovers** to unaffected stores in the same chain.
- ▶ So chains price uniformly with respect to **demand** but respond locally to **costs**. Uniform pricing is not the explanation for the spirits results; price points and the discreteness of retail pricing are.
- ▶ Practical upshots: (a) local tax changes are clean cost shifters for the supply side of a demand model (the Conlon Rao supply moments); (b) the elasticity you back out from a tax change is the elasticity of the stores that actually changed price, with the price-point weighting from the previous slide.

Some challenges

- ▶ Pless and van Benthem (AEJ:Applied 2019): use pass-through > 1 as a test for market power? Only if you are willing to rule out convex demand under competition, and to assume constant marginal cost.
- ▶ Chetty, Looney, Kroft (AER 2009)

$$\log q(p, \tau^S) = \alpha + \beta \log p + \theta_\tau \beta \log(1 + \tau^S)$$
$$\theta_\tau = \frac{\partial \log q}{\partial \log(1 + \tau^S)} / \frac{\partial \log q}{\partial \log p} = \frac{\varepsilon_{q, 1 + \tau^s}}{\varepsilon_{q, p}}$$

- ▶ Use θ as evidence of “tax salience” (is it?). With imperfect competition the posted price responds to the tax too, so the incidence and excess-burden formulas change: Kroft, Laliberté, Leal-Vizcaíno, Notowidigdo (ReStud 2024) derive them with salience and market power together, find near-complete pass-through of sales taxes, consumers who respond more to prices than to taxes, and a marginal excess burden larger than the standard formulas give.
- ▶ Can we identify markups from pass-through alone...?!? Only jointly with curvature. Which is where we started.